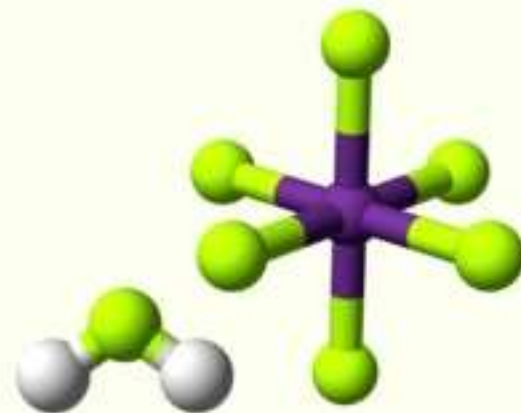


CHEMISTRY

CARBON & ITS COMPOUNDS

MODULE - 12

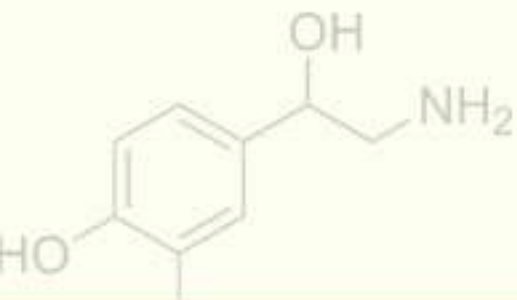
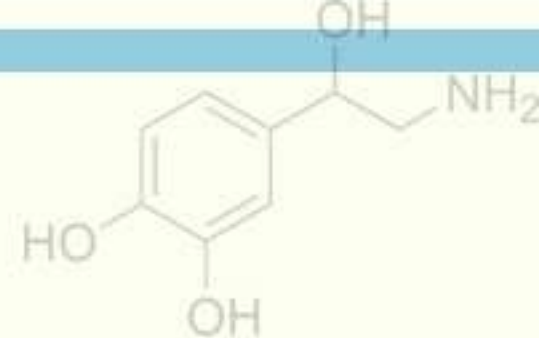
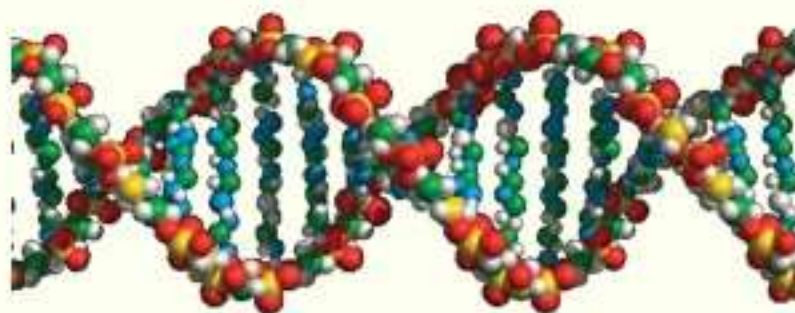


COMPOUNDS

INORGANIC COMPOUNDS



ORGANIC COMPOUNDS



WHAT ARE LIVING THINGS MADE UP OF?



Organisms



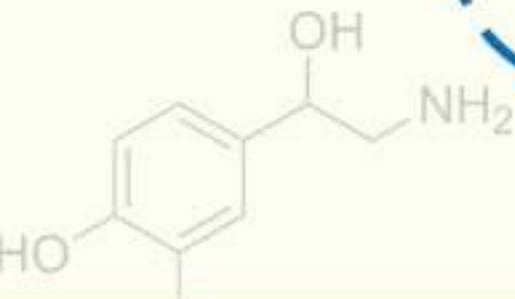
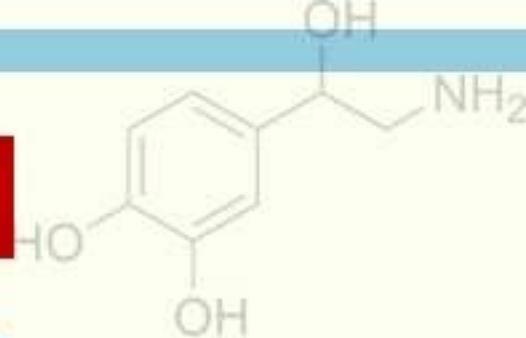
Cells



macromolecules



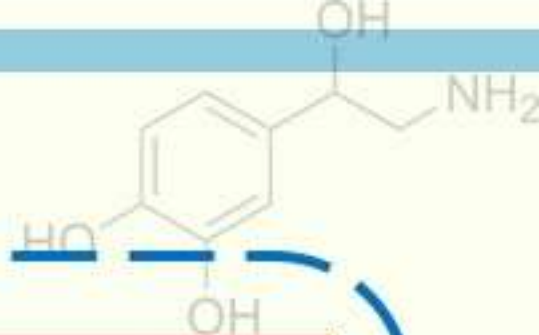
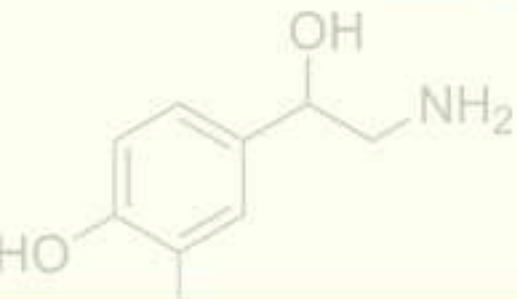
Organic compounds



CARBON



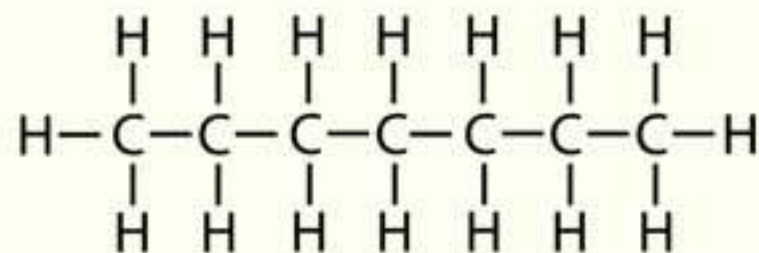
- Carbon (from Latin: carbo "coal") is a chemical element with symbol C and atomic number 6.
- It belongs to group 14 of periodic table.
- It is nonmetallic in nature.
- Carbon is tetravalent—making four electrons available to form covalent chemical bonds.



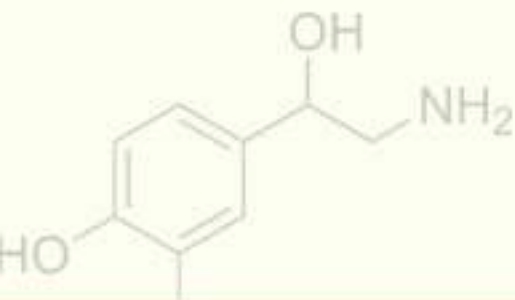
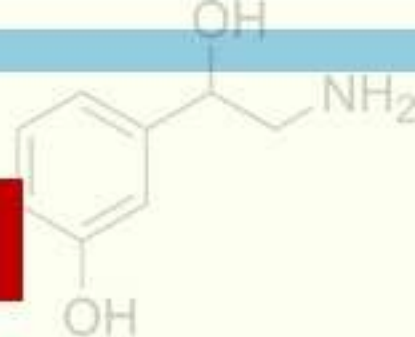
POSITION OF CARBON IN THE PERIODIC TABLE

1	1 H	POSITION OF CARBON IN THE PERIODIC TABLE																2 He						
2	3 Li	4 Be																	5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg																	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr						
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe						
6	55 Cs	56 Ba	57 La	* 72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn						
7	87 Fr	88 Ra	89 Ac	* 104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og						
				* 58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu							
				* 90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr							

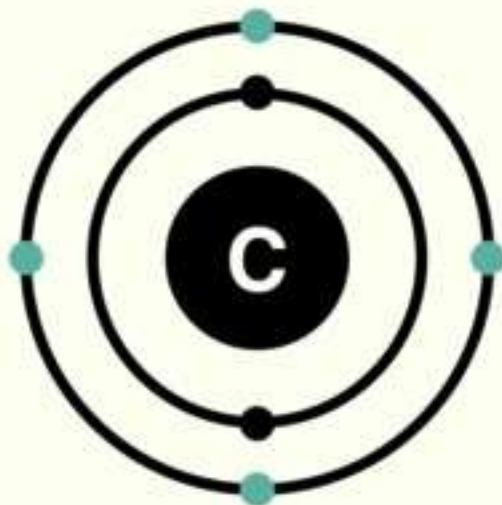
WHAT MAKES CARBON UNIQUE IN NATURE?



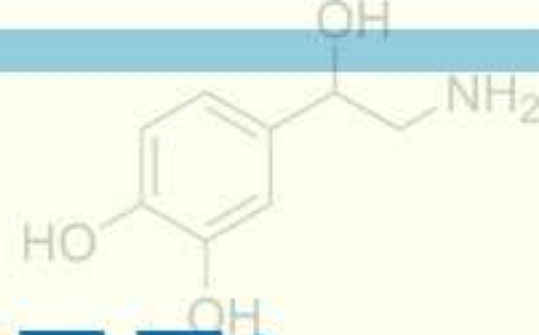
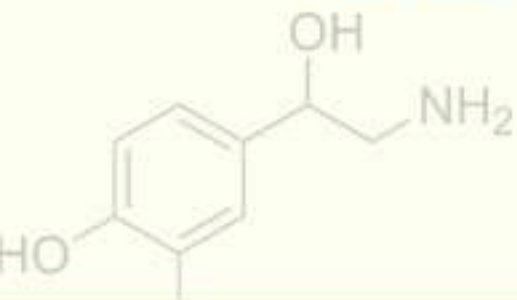
- **Covalent bonding**- carbon can share electrons to four other elements including another carbon.
- **Catenation**- carbon has a unique property to form long chains by forming covalent bonds
- **Allotropy**- carbon can be found in different forms due to its valency.

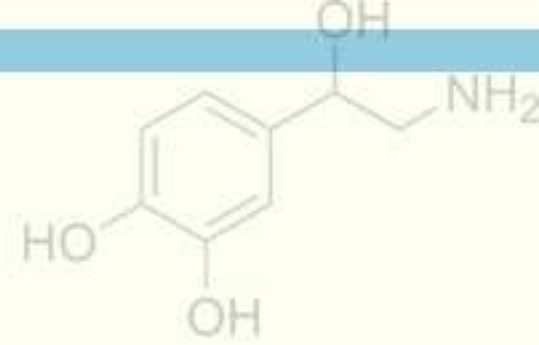


TETRAVALENCY OF CARBON



- Carbon has 4 electrons in its outer orbit.
- It can neither donate nor accept 4 electrons to complete the octet.
- Carbon can share the electrons to form different compounds.





ALLOTROPES OF CARBON

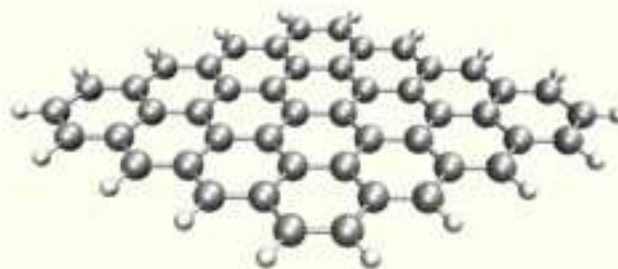
DIAMOND



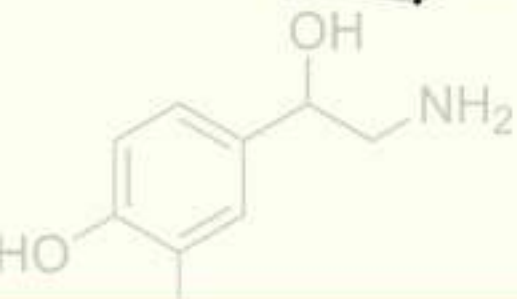
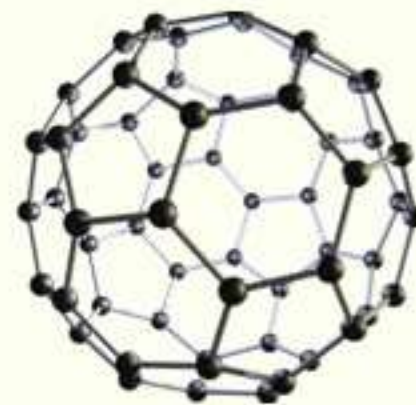
GRAPHITE



GRAPHENE



BUCKMINSTER FULLERENE



ALLOTROPES OF CARBON



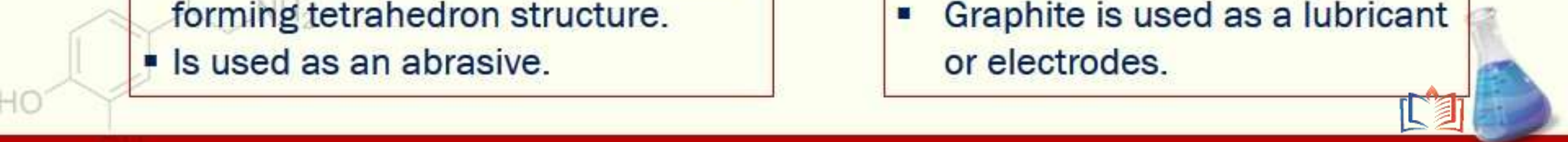
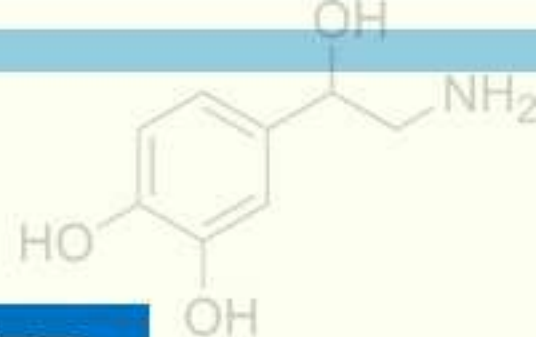
DIAMOND

- Diamond is shiny and transparent.
- Diamond is a poor conductor of electricity but a good conductor of heat.
- Diamond is the hardest substance known in nature.
- Diamond is more dense
- Each C atom in diamond is covalently bonded with 4 C atoms, forming tetrahedron structure.
- Is used as an abrasive.



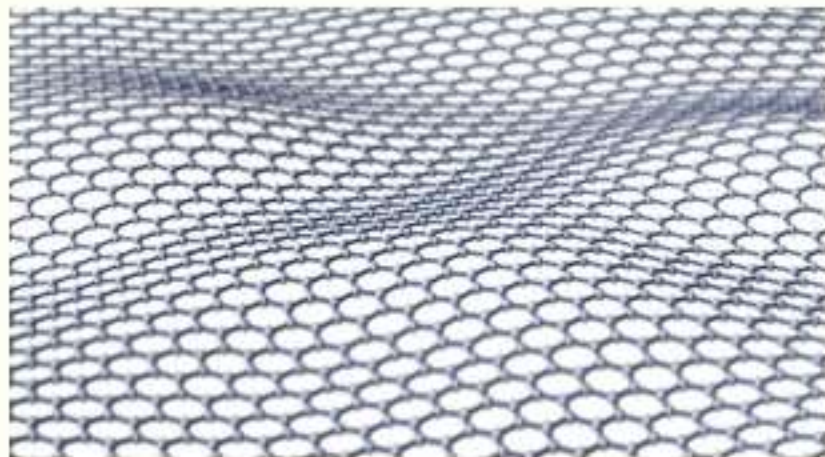
GRAPHITE

- Graphite is dull and opaque.
- Graphite is a good conductor of heat and electricity.
- Graphite is soft and slippery to touch.
- Graphite is less dense
- Each C atom is covalently bonded to 3 other C atoms in a hexagonal structure.
- Graphite is used as a lubricant or electrodes.

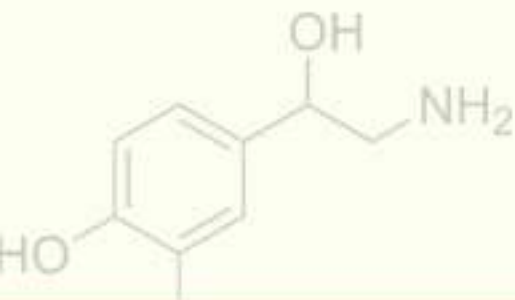
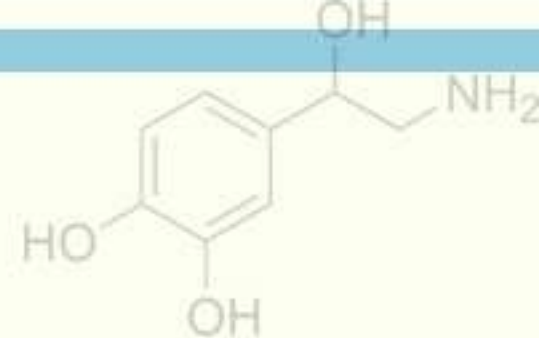


ALLOTROPE OF CARBON

GRAPHENE



- An allotrope of carbon consisting of a single layer of carbon atoms arranged in a hexagonal lattice.
- It is the strongest material ever tested,
- efficiently conducts heat and electricity with very little resistant.
- It is nearly transparent
- Graphene can be used as a chemical sensors, LED lighting
- In Solar Cell battery supercapacitor.

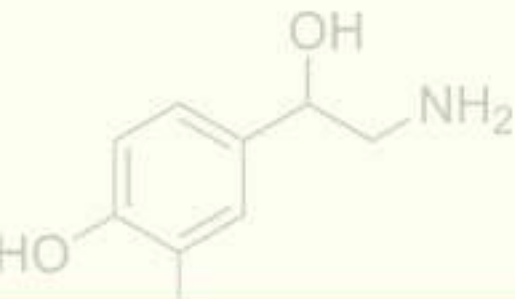
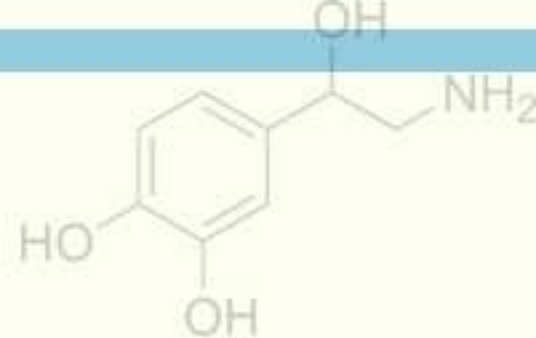


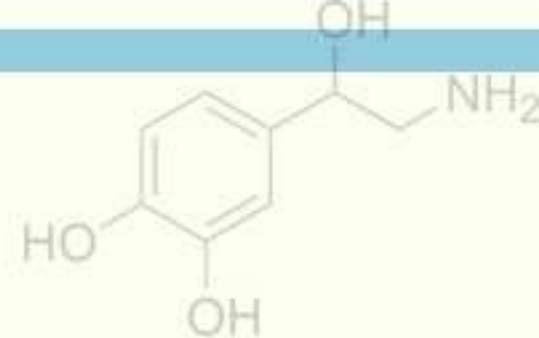
ALLOTROPES OF CARBON

FULLERENE



- Fullerenes or Bucky-balls were discovered in 1985.
- Fullerenes are used as super conductors
- In electronic and microelectronic devices
- In optical devices
- In batteries as charge conies
- In medical industry to bind specific antibodies to target resistant bacteria and cancer cells.

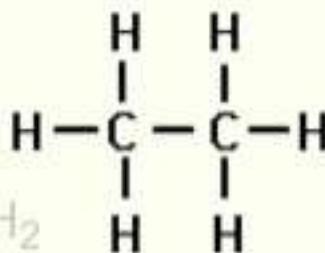




CARBON COMPOUNDS

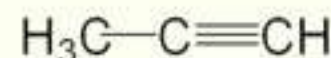
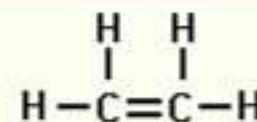
SATURATED

- Each Carbon has a single bond.
- No more electron is available to the atoms for sharing.



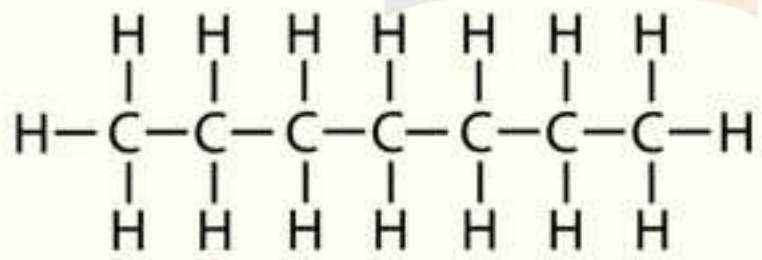
UNSATURATED

- Two or more than two Carbon will have double or triple bond.
- One or two pairs of electrons are available for sharing.

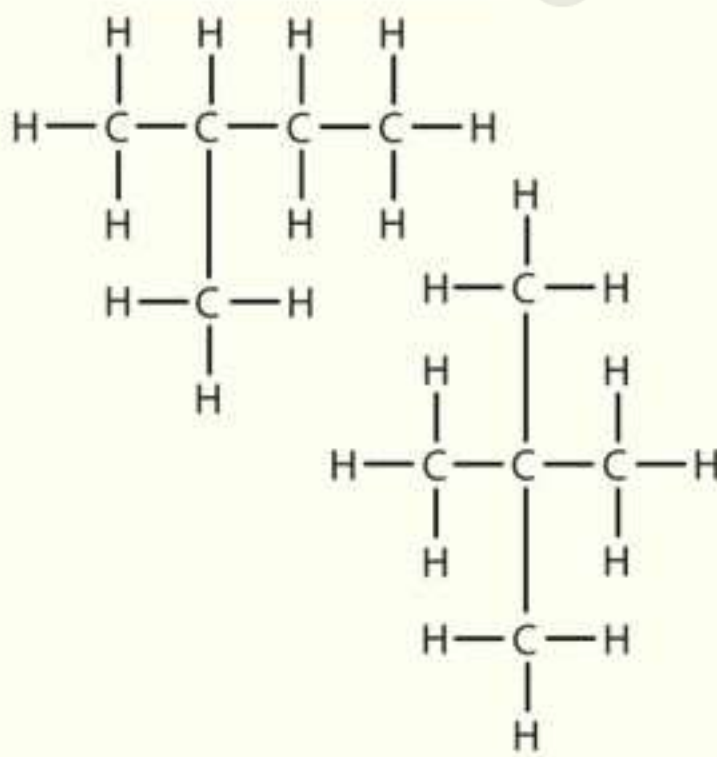


CARBON COMPOUNDS

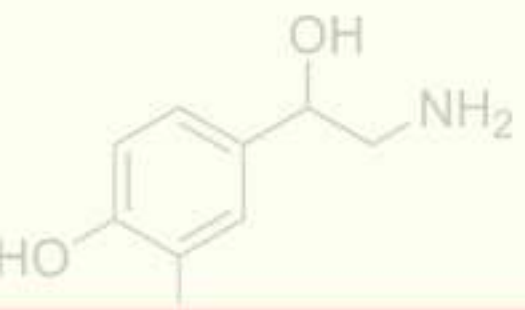
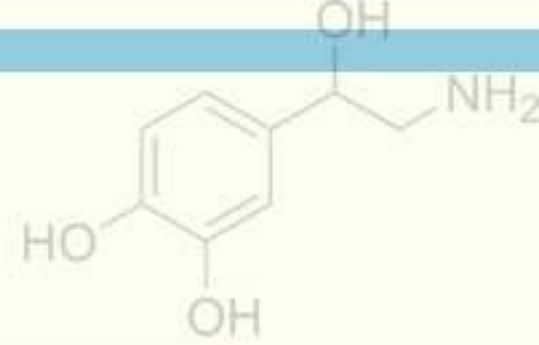
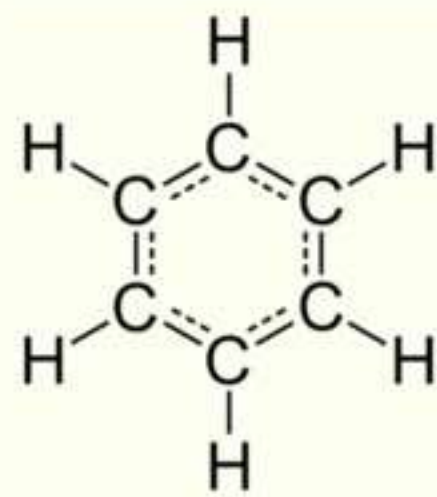
STRAIGHT CHAIN



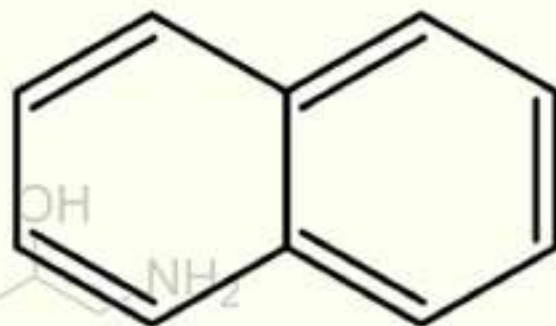
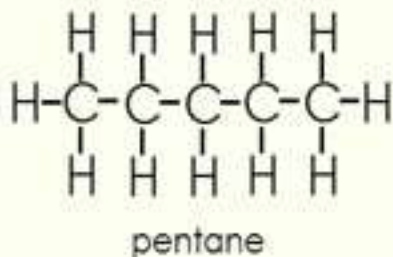
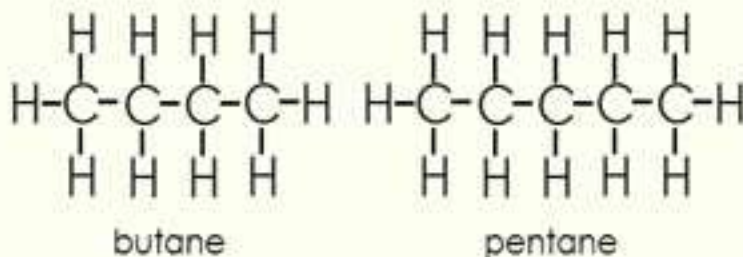
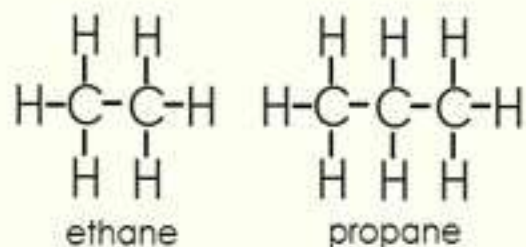
BRANCHED CHAIN



CLOSED CHAIN/RING



HYDROCARBONS



Naphthalene

- A hydrocarbon is an organic compound made of nothing more than carbons and hydrogen.
- **Saturated hydrocarbons** have as many hydrogen atoms as possible attached to every carbon
- **Unsaturated hydrocarbons** have double and/or triple bonds between some of the carbon atoms.
- **Aliphatic hydrocarbons** are compounds of hydrogen and carbon that do not contain benzene rings
- **Aromatic hydrocarbons** are derived from/structure of the benzene rings.



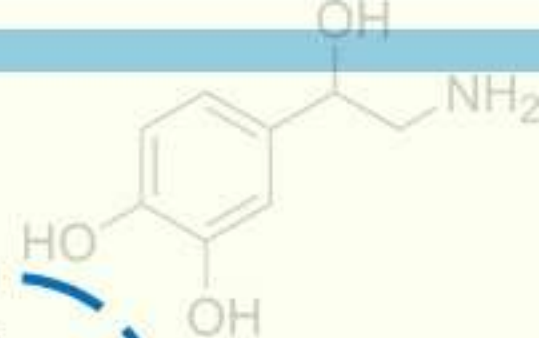
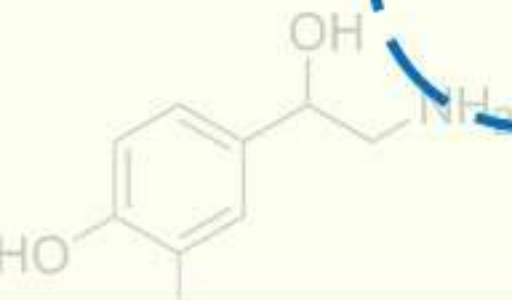
HYDROCARBONS



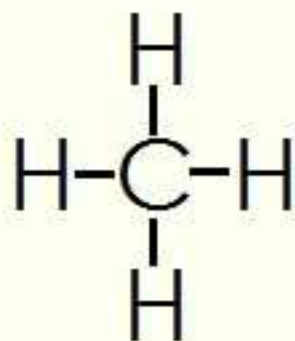
Saturated hydrocarbons
will generally give a
clean blue flame



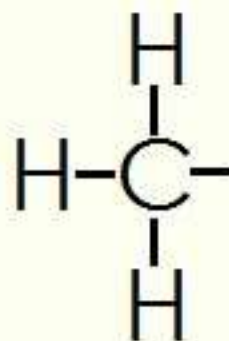
Unsaturated carbon
compounds will give a
yellow flame with lots of
black smoke.



FUNCTIONAL GROUP



methane



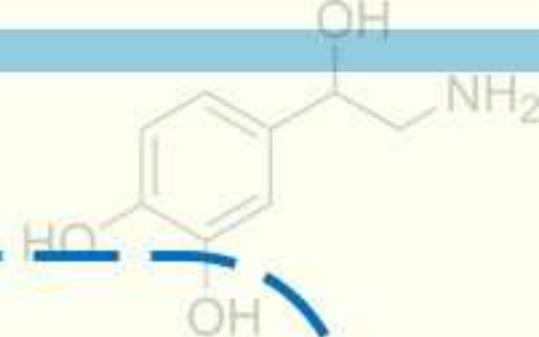
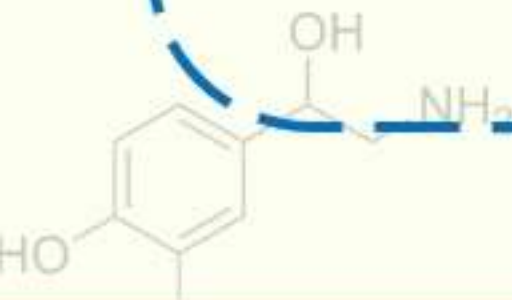
methyl group

1

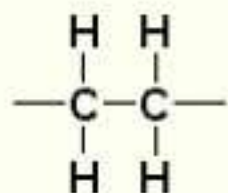
It is the atom or group of atoms present in the molecule which determines the characteristics property of organic compounds.

2

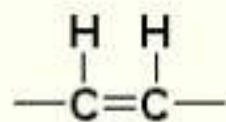
If one hydrogen(H) in methane CH_4 is replaced by an $-\text{Cl}$ group, then a compound CH_3Cl is formed. This Cl group is the functional group



FUNCTIONAL GROUPS IN CARBON COMPOUNDS



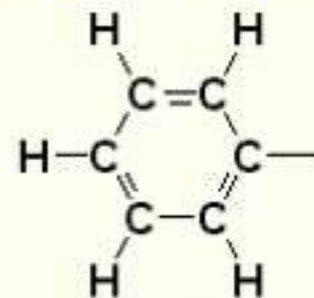
alkane



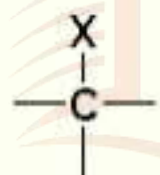
alkene



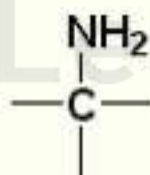
alkyne



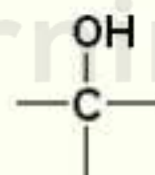
phenyl



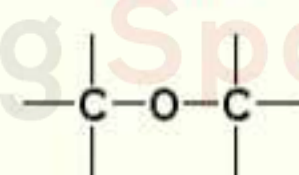
alkyl halide
(X = F, Cl, Br, I)



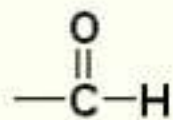
amine



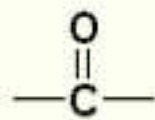
alcohol



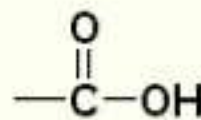
ether



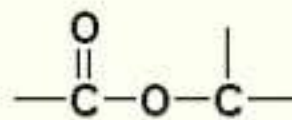
aldehyde



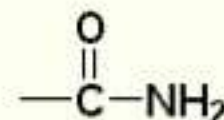
ketone



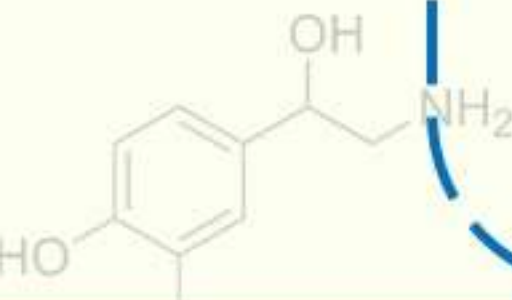
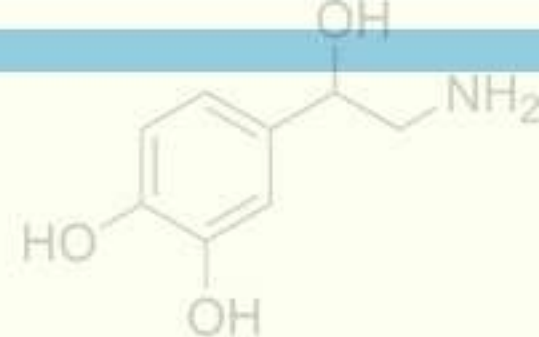
carboxylic acid



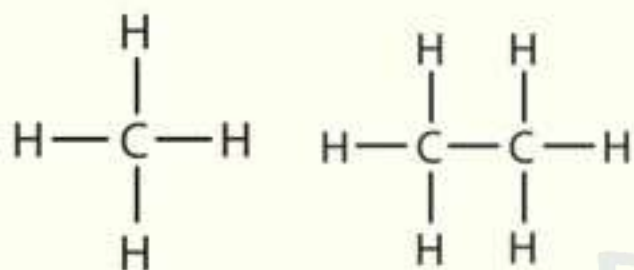
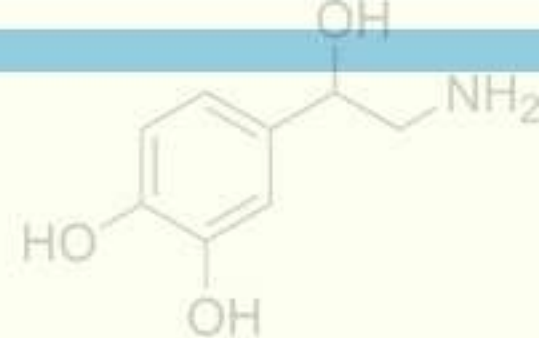
ester



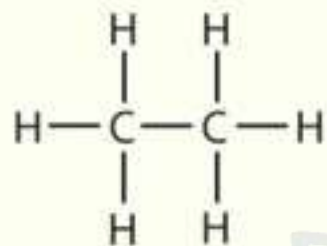
amide



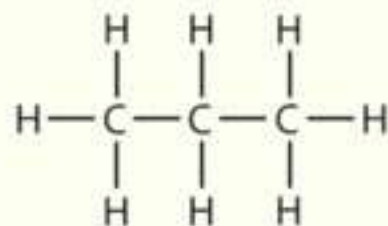
HOMOLOGOUS SERIES



Methane



Ethane

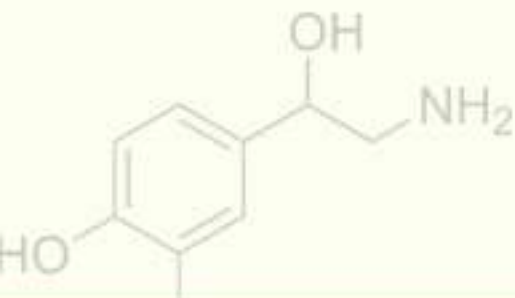


Propane

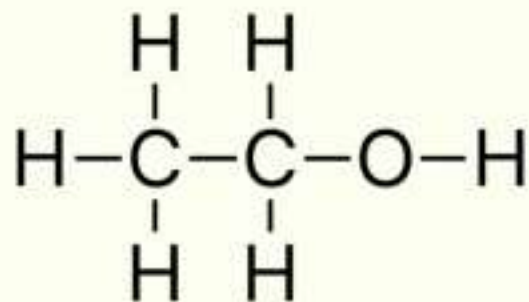
A series of compounds in which the same functional group substitutes **hydrogen** in a carbon chain called the homologous series.

Eg. Alkanes are a homologous series

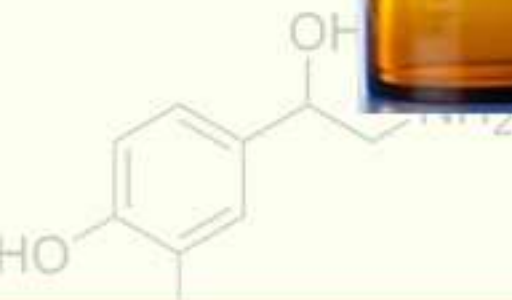
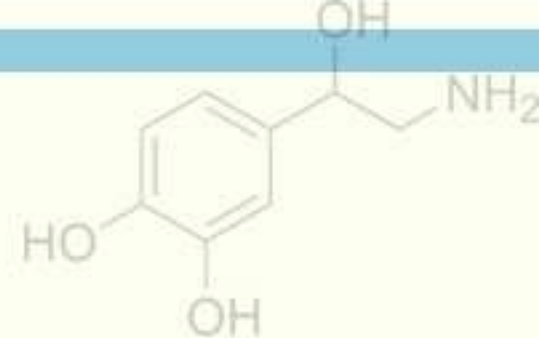
General formula $\text{C}_n \text{H}_{2n+2}$



ETHANOL



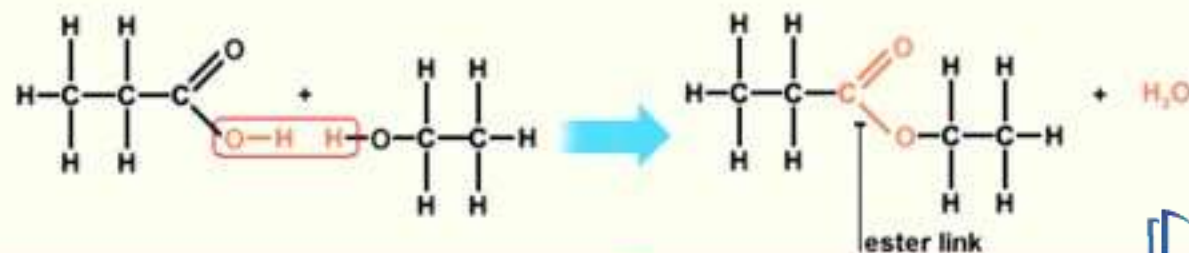
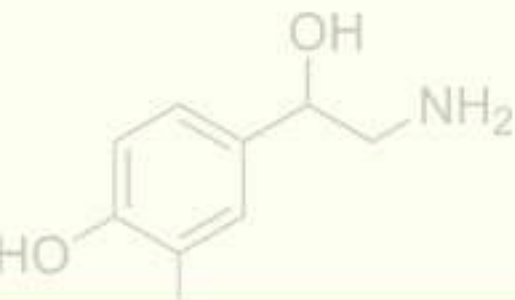
- Ethanol is an alcohol and is the active ingredient of all alcoholic drinks.
- Ethanol is produced by fermentation of carbohydrates.
- Ethanol is also soluble in water in all proportions.
- Ethanol is a very good solvent, it is used in medicines such as tincture iodine, cough syrups, and many tonic.



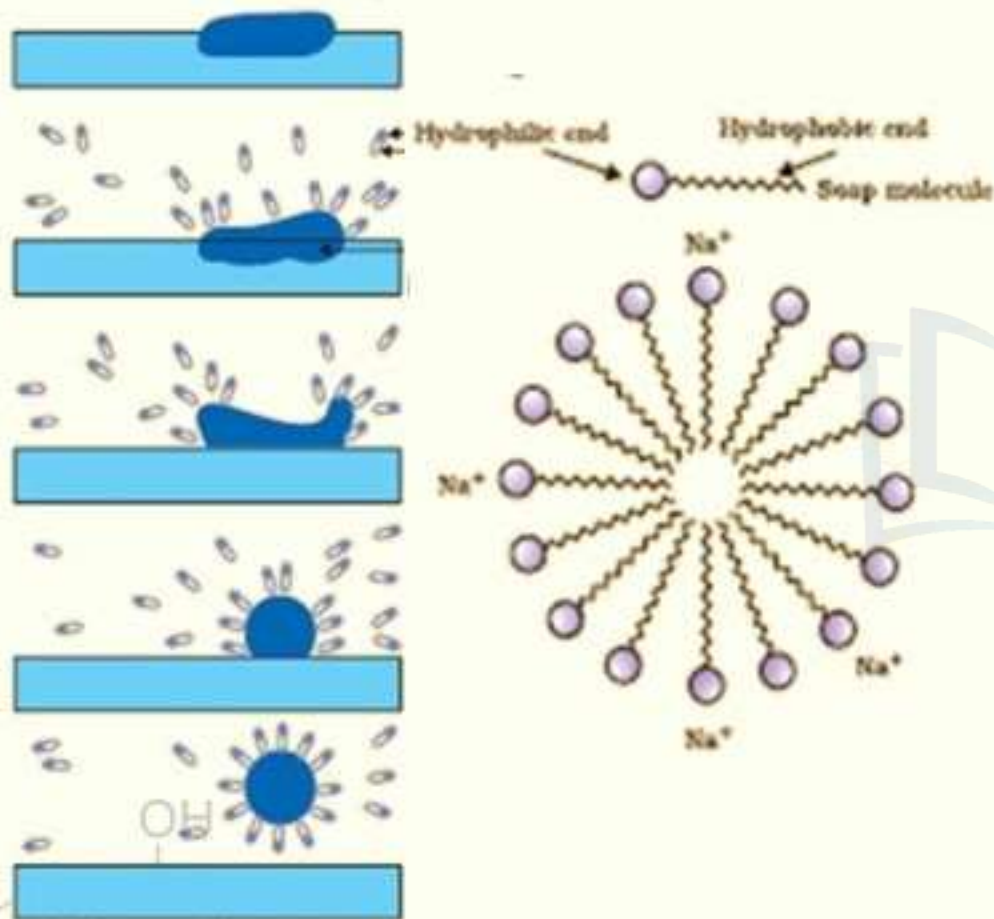
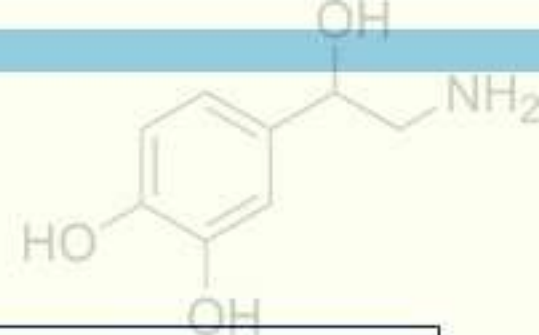
ESTERS



- When an alcohol reacts with a carboxylic acid, it forms an ester.
- Esters are sweet-smelling substances.
- These are used in making perfumes and as flavoring agents.
- Esters react in the presence of an acid or a base to give back the alcohol and carboxylic acid.
- This reaction is known as saponification because it is used in the preparation of soap.



DETERGENTS



- Detergents are generally ammonium or sulphonate salts of long chain carboxylic acids.
- The long chain carboxylic acid is the water repelling/hydrophobic end.
- The polar ammonium and sulphonate ions becomes the hydrophilic end.
- The detergent molecule form a micelle structure to grab a dirt/grease molecule from the clothes.

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